Fidel Omar Tito Cruz

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Education

University of Central Florida (UCF), Orlando - Florida *Aug 2025 - Present*

Current MS student in Computer Vision · Fulbright Scholarship

Universidad Nacional de Ingeniería (UNI), Lima - Perú *Aug 2016 - Jul 2021*

B.S. Mechatronics Engineering. Scholarship student - Permanencia de Estudios PRONABEC

Research Interests

Large Language Models, Vision-Language Models, Natural Language Processing, Multimodal Learning, Model Alignment and Post-Training, Computer Vision, Deployable ML Systems.

Professional Experience

Adobe - ML Engineer Intern - San Jose, California *May 2026 - Aug 2026*

- Improved the recommendation system behind the Adobe Acrobat AI Assistant, targeting the CTR of the next-question recommendations displayed to users.
- Trained and evaluated multiple ML models and LLM-based approaches to generate and rank recommendations, comparing them against the existing production baseline.
- Delivered an estimated 16% relative increase in CTR on AI Assistant recommendations.

QUBIKA - ML Engineer Senior - Austin, Texas *Aug 2024 - Aug 2025*

- Developed and deployed a text-to-code LLM system with Retrieval-Augmented Generation (RAG), and large language models (OpenAI), achieving a 90%+ accuracy and producing a commercial version available to be offered as a product to external clients.
- Built and deployed an AI document parser using AWS, OpenSearch, RAG, Claude, Gemini VLM capabilities, and OCR, designing a multi-step pipeline to extract, rank, and parse PDF content, enabling accurate extraction of financial data.
- Developed an internal framework in Databricks for deploying traditional ML models with standardized pipelines for preprocessing, training, and validation, used to build attrition, look-alike, and cross-selling models for a financial client.

INTERCORP - ML Engineer - Lima, Perú *Sep 2022 - Jul 2024*

- Worked on an internal MLOps framework for a consulting company servicing retail industry using GCP, Kubeflow, Vertex AI, and Bash; reduced deployment time from several days to a single day, and improved efficiency for data scientists by eliminating manual configurations.
- Developed and deployed a Text-to-SQL LLM system leveraging GCP, OpenAI API, RAG, and SQL stack to answer simple internal company questions about business metrics without requiring manual SQL querying, enabling internal users to receive rapid, automated SQL responses to common analytical questions.
- Optimized an accurate incremental sales calculation algorithm for big data (10M rows/month), facilitating faster comparison testing between groups using linear programming, optimizing resource consumption.

LEASEIN - Data Scientist - Lima, Perú *Oct 2021 - Aug 2022*

- Built an automated client-scoring model to set laptop rental conditions, preprocessing an imbalanced dataset and training with XGBoost, Python, and scikit-learn.
- Automated resume screening with NLP and built reporting, massive email/WhatsApp messaging, and RPA data-extraction programs in Python and JavaScript.

Research Experience

PUBLICATIONS

[1] **F. Tito***, N. Redmond*, D. Sharma, S. Azad, S. Mitra, S. Vyas, Y. S. Rawat, "MaRiO: Multi-agent Collaborative Reasoning via Shared Observations in MLLMs". The Fortieth Annual Conference on Neural Information Processing Systems (NeurIPS) 2026

[2] **F. Tito**, A. Sanchez, S. Farfan, G. Bejarano, "SignRR: Retrieve and Refine Real Motion for Sign Language Production". 37th British Machine Vision Conference (BMVC), 2026. Oral presentation.

[3] **F. Tito**, N. Ghafouri, Z. Wang, P. Khosravi, Y. Tian, C. Chen, "Longitudinal 3D Foundation Modeling for Neoadjuvant Breast Cancer Response Prediction from Serial DCE-MRI". Applications of Medical AI (AMAI) Workshop at MICCAI, 2026. Oral presentation.

[4] A. Sanchez, **F. Tito**, J. Ochoa, "MassFlow: R3D Flow-Guided Mass Estimation Model". Proceedings of the LatinX in AI Workshop at the International Conference on Machine Learning (ICML), 2025, Vancouver, Canada.

[5] **F. Tito**, G. Bejarano, "Generative Interpolation of Sign Language Poses using RVQ-VAE". Proceedings of the LatinX in AI Workshop at the Neural Information Processing Systems Conference (NeurIPS) 2024, Vancouver, Canada.

[6] **F. Tito**, R. Doloriet, A. Guardia, J. Francia, "A comparison of Gain Scheduling PID and μ-Synthesis Robust Level Control for a Conical Tank System". Proceedings of the IEEE XXVIII International Conference on Electronics, Electrical Engineering and Computing (INTERCON) 2021, Lima, Perú.

[7] M. Nekoui, **F. Tito**, L. Cheng, "EAGLE-Eye: Extreme-pose Action Grader using detaiL bird's-Eye view". Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) 2021, Waikoloa, Hawaii, USA.

[8] M. Nekoui, **F. Tito**, L. Cheng, "FALCONS: FAst Learner-grader for CONtorted poses in Sports". Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops, 2020, Washington USA.

ONGOING RESEARCH PROJECTS

Recovering a Pruned and Quantized MLLM with Its Own Small Parameters *Sep 2026 - Current*

A new method to recover accuracy by training only MLLM small parameters, leaving every 4-bit compressed LoRA weights untouched as we add quantization and structural pruning on attention and NN layers. Working at the University of Central Florida.

RESEARCH LABS

Center for Research in Computer Vision - Prof. Yogesh Rawat *Oct 2025 - Current*

University of Central Florida (UCF)
Constructing a multi-view spatial-reasoning benchmark to show challenging cases where MLLMs (e.g Qwen 2.5, LLaVA 1.5, etc) fail to answer spatial questions accurately.
Keywords: MLLM/VLM evaluation, multi-agent views, spatial QA, Hugging Face, Slurm, CUDA

Vision and Learning Lab - Canada - Prof. Li Cheng *Jan 2020 - Apr 2020*

University of Alberta Research Experience Program (UARE) - Internship
Researched pose estimation for extreme poses, focusing on robust detection under challenging body configurations.
Keywords: Pose estimation, PyTorch, object detection, ST-GCN, image processing.

Achievements, Honors and Awards

- Selected by YC Startup School *Jul 2026*
- Fulbright scholarship for graduate studies, given by Fulbright *Jun 2024*
- Financial award for research project, given by Vice-Rectorate Research UNI *May 2021*
- Selected by University of Alberta Research Experience Program (UARE) internship *Oct 2019*
- 2nd place, CSpace International Competition 2019 France - CANSAT Category. *Jul 2019*
- Scholarship - Beca Permanencia de Estudios 2018 given by PRONABEC *Sep 2018*

Skills

Python, C++, HTML, CSS, JavaScript, PyTorch, TensorFlow, Hugging Face (Transformers, Datasets, Accelerate), LoRA / PEFT, CUDA, Slurm, Weights & Biases, OpenCV, scikit-learn, React, ExpressJS, SQL, Big Data, Google Cloud Platform (GCP), Amazon Web Services (AWS), Azure, Bigquery, Kubeflow, MLflow, Docker, Git, Kubernetes, LangChain, MongoDB, MATLAB, Linux, Computer Vision, NLP, ML, Statistics, LLM, VLM, LaTeX

Certifications

Google Certified Associate Cloud Engineer *Jun 2025*

Databricks Certified Machine Learning Associate *Oct 2024*

Languages

- English, Advanced Level - TOEFL iBT 97/120, TOEFL iBT MyBest 101/120
- Spanish, Native

Extra-curricular Activities

- Mentee on LatinX in AI Mentoring Program. Current member of LatinX in AI Community
- Vice Chair IEEE EMBS UNI 2020
- Volunteer at InspírateUNI 2017: Vocational orientation program for UNI applicants